

RESEARCH PROPOSAL SEMINAR

On the Complexity of Fixed Confidence Sequential Hypothesis Testing with HMM Observations

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Doctoral Committee

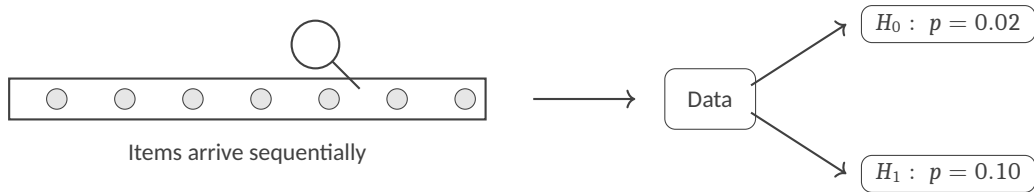
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The Setting

- A factory produces items continuously on a conveyor belt.
- A machine responsible for production has recently been recalibrated.
- We would like to determine whether the machine is operating correctly or producing too many defective items.
- Two possibilities are considered:
 - H_0 : the defect rate is at the acceptable level (e.g., $p = 0.02$),
 - H_1 : the defect rate is too high (e.g., $p = 0.10$).
- Items arrive one at a time, and production does not have a natural endpoint.
- After inspecting each item, we must decide whether to
 - stop and declare H_0 ,
 - stop and declare H_1 , or
 - inspect one more item.

Sequential Hypothesis Testing

- The production-line example is an instance of a *sequential hypothesis testing* problem.
- Observations arrive one at a time, and after each observation, we decide whether to stop or continue.
- There are two competing hypotheses:
 - H_0 : acceptable defect rate,
 - H_1 : excessive defect rate.



Policy for Sequential Hypothesis Testing

A policy π consists of two components:

- A **stopping rule**

which decides whether to *continue* or *stop* after observing $(X_1, \dots, X_t)$.

$$\tau = \inf\{t : \text{stop at time } t\},$$

- A **decision rule**

$$\hat{H} \in \{H_0, H_1\},$$

which selects a hypothesis when the test stops.

Requirements

A valid policy must satisfy:

1. **Almost sure termination**

$$\mathbb{P}_{H_i}(\tau < \infty) = 1, \quad \text{where } i \in \{0, 1\}$$

$$\pi : (X_1, X_2, \dots) \longrightarrow (\tau, \hat{H})$$

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Requirements

A valid policy must satisfy:

1. **Almost sure termination**

$$\mathbb{P}_{H_i}(\tau < \infty) = 1, \quad \text{where } i \in \{0, 1\}$$

2. **Error constraint**

$$\mathbb{P}_{H_0}(\hat{H} = H_1) \leq \delta, \quad \mathbb{P}_{H_1}(\hat{H} = H_0) \leq \delta,$$

where δ is the prescribed error tolerance.

Many Policies are Possible

- There is no unique way to design a sequential testing procedure.
- Different policies may use very different numbers of observations before making a decision.
- Some policies stop very quickly but make more errors.
- Others are more conservative and require many observations.

δ -Correct Policies

$$\Pi_\delta := \left\{ \pi = (X_1, X_2, \dots) \implies (\tau, \hat{H}) : \mathbb{P}_{H_i}(\tau < \infty) = 1, \mathbb{P}_{H_i}(\hat{H} \neq H_i) \leq \delta, \text{ where } i \in \{0, 1\} \right\}.$$

Stopping Time

- Among all admissible policies, we would like to find one that reaches a decision as quickly as possible while respecting prescribed error constraints.
- Since the number of observations required is random, the stopping time itself is a random variable.
- Formally,

$$\tau = \inf\{n : \text{the policy decides to stop at time } n\}.$$

- The decision to stop at time n depends only on observations seen up to time n .
- Consequently, different sample paths may lead to different values of τ .

Expected Stopping Time

- A good sequential procedure should make decisions quickly while controlling the probabilities of error.
- Expected Stopping Time is also called as **Sample Complexity** of the policy

Objective

Among all δ -correct sequential tests, find the policy that minimizes the expected stopping time:

$$\inf_{\pi \in \Pi_\delta} \mathbb{E}_{H_i}[\tau] \quad \text{where } i \in \{0, 1\}$$

General Information-Theoretic Lower Bound

Consider the simple hypotheses

$$H_0 : \mathbb{P} = \mathbb{P}_0 \quad \text{vs} \quad H_1 : \mathbb{P} = \mathbb{P}_1,$$

based on a sequence of observations

$$X_1, X_2, \dots$$

where $X_1, X_2, \dots \sim \mathbb{P}$.

Here $\mathbb{P}$ is a probability law and is unknown. Assume H_1 Hypothesis is true.

Lower Bound

For any δ -correct sequential test,

$$\mathbb{E}_{H_1}[\tau] \geq \frac{d(\delta, 1 - \delta)}{D_{\text{KL}}(\mathbb{P}_1 \parallel \mathbb{P}_0)}$$

where $d(\cdot, \cdot)$ denotes the Bernoulli KL Divergence.

- **Numerator:** The term $d(\delta, 1 - \delta)$ arises from a *change-of-measure argument*. Applying the *data-processing inequality* to the final decision variable gives

$$\mathbb{E}_{H_1}[\log L_\tau] \geq d(\delta, 1 - \delta).$$

where

$$L_\tau = \frac{\mathbb{P}_1(X_{1:\tau})}{\mathbb{P}_0(X_{1:\tau})}.$$

- **Denominator:** The quantity $D_{\text{KL}}(\mathbb{P}_1 \parallel \mathbb{P}_0)$ is the KL Divergence rate. It appears via

$$\mathbb{E}_{H_1}[\log L_\tau] = D_{\text{KL}}(\mathbb{P}_1 \parallel \mathbb{P}_0) \mathbb{E}_{H_1}[\tau].$$

Sequential Hypothesis Testing of IID Samples

Observe i.i.d. samples where P is unknown

$$X_1, X_2, \dots \sim P$$

and consider the simple hypotheses

$$H_0 : P = P_0, \quad H_1 : P = P_1.$$

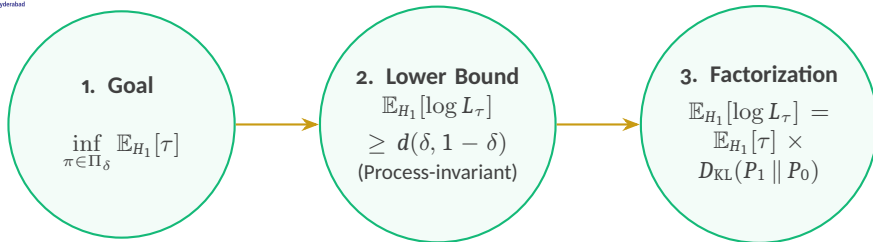
We require the test to satisfy a prescribed Type-I and Type- II error constraint

$$\mathbb{P}_{H_0}(\hat{H} = H_1) \leq \delta. \quad \mathbb{P}_{H_1}(\hat{H} = H_0) \leq \delta.$$

Our objective is to minimize the expected number of samples :

$$\boxed{\inf_{(\pi \in \Pi_\delta)} \mathbb{E}_{H_i}[\tau]} \quad \text{where } i \in \{0, 1\}$$

The Lower Bound Logic



- **The Core Theoretical Mission:** The second block holds universally across all environments. The primary focus of us is to establish the third block in iid case

Information Processing Lemma

Kaufmann E., Cappé O., and Garivier A. (2016). "On the Complexity of Best-Arm Identification in Multi-Armed Bandit Models." *Journal of Machine Learning Research*, 17(1), 1–42. (Lemma 1.)

Extending the Lower Bound: The Role of Wald's Lemma

- For iid the joint distribution splits into products:

$$L_\tau = \frac{P_1(X_1, X_2, \dots, X_\tau)}{P_0(X_1, X_2, \dots, X_\tau)} = \prod_{t=1}^{\tau} \frac{P_1(X_t)}{P_0(X_t)}$$

- Recovering Additivity via the Logarithm:

$$\log L_\tau = \log \left(\prod_{t=1}^{\tau} \frac{P_1(X_t)}{P_0(X_t)} \right) = \sum_{t=1}^{\tau} \log \frac{P_1(X_t)}{P_0(X_t)}$$

- Deriving the Denominator via Wald's Lemma:

$$\mathbb{E}_{H_1}[\log L_\tau] = \mathbb{E}_{H_1} \left[\sum_{t=1}^{\tau} \log \frac{P_1(X_t)}{P_0(X_t)} \right] = \mathbb{E}_{H_1}[\tau] \cdot D_{\text{KL}}(P_1 \parallel P_0).$$

This factorization isolates $\mathbb{E}_{H_1}[\tau]$ and KL Divergence term.

Foundational Reference

SPRT and Fundamental Lower Bound

Algorithm 1 Sequential Probability Ratio Test

```
1: Input: error tolerance  $\delta$ 
2: Set threshold  $A = \log(1/\delta)$ 
3:  $n \leftarrow 0$ 
4: while true do
5:    $n \leftarrow n + 1$ 
6:   Observe  $X_n$ 
7:   Compute  $L_n = \sum_{t=1}^n \log \frac{P_1(X_t)}{P_0(X_t)}$ 
8:   if  $L_n \geq A$  then
9:     Stop and declare  $H_1$ 
10:  end if
11:  if  $L_n \leq -A$  then
12:    Stop and declare  $H_0$ 
13:  end if
14: end while
```

Performance of SPRT

Assuming H_1 is true, the expected sample path satisfies:

$$\mathbb{E}_{H_1}[\tau_{\text{SPRT}}] \leq \frac{\log(1/\delta)}{D_{\text{KL}}(P_1 \| P_0)} (1 + o(1)).$$

Theoretical Properties:

1. **δ -PAC Classification:** By designing the termination boundaries to grow strictly on the scale of $\mathcal{O}(\log(1/\delta))$.
2. **Asymptotic Convergence:** This analytical upper bound follows strictly from the *Strong Law of Large Numbers* (SLLN)

Asymptotic Optimality in the i.i.d. Case

We establish the exact match between our algorithmic performance and the theoretical threshold as $\delta \rightarrow 0$:

- **Step 1: The Asymptotic Lower Bound (for any $\pi \in \Pi_\delta$)**

$$\mathbb{E}_{H_1}[\tau] \geq \frac{d(\delta, 1 - \delta)}{D_{\text{KL}}(P_1 \parallel P_0)} \xrightarrow{\delta \rightarrow 0} \frac{\log(1/\delta)}{D_{\text{KL}}(P_1 \parallel P_0)} (1 - o(1))$$

- **Step 2: The Algorithmic Upper Bound (SPRT Performance)**

$$\mathbb{E}_{H_1}[\tau_{\text{SPRT}}] \leq \frac{\log(1/\delta)}{D_{\text{KL}}(P_1 \parallel P_0)} (1 + o(1))$$

- **Step 3: The Asymptotic Convergence as $\delta \rightarrow 0$**

$$\lim_{\delta \rightarrow 0} \frac{\mathbb{E}_{H_1}[\tau_{\text{SPRT}}]}{\log(1/\delta)} = \frac{1}{D_{\text{KL}}(P_1 \parallel P_0)}$$

Sequential Hypothesis Testing of Markov Samples

Observe a sequence of Markov-dependent samples where the transition dynamics are unknown

$$X_1, X_2, \dots \sim P \quad , \text{where } P \text{ is unknown TPM}$$

$$P(X_{t+1} \mid X_t, X_{t-1}, \dots) = P(X_{t+1} \mid X_t)$$

and consider the simple hypotheses on the transition probability matrices (TPMs)

$$H_0 : P = P^{(0)}, \quad H_1 : P = P^{(1)}.$$

We require the test to satisfy a prescribed Type-I and Type-II error constraint

$$\mathbb{P}_{H_0}(\hat{H} = H_1) \leq \delta, \quad \mathbb{P}_{H_1}(\hat{H} = H_0) \leq \delta.$$

Our objective remains to minimize the expected number of samples (Average Sample Number):

$$\boxed{\inf_{(\pi \in \Pi_\delta)} \mathbb{E}_{H_i}[\tau]} \quad \text{where } i \in \{0, 1\}$$

Extending the Lower Bound: Likelihood Decomposition

- **Decomposition via Transition Dynamics:** For Markov-dependent observations, the joint distribution splits into an initial distribution and conditional transition matrix products:

$$L_\tau = \frac{\mathbb{P}_{H_1}(X_1, \dots, X_\tau)}{\mathbb{P}_{H_0}(X_1, \dots, X_\tau)} = \frac{\mu_{P^{(1)}}(X_1)}{\mu_{P^{(0)}}(X_1)} \prod_{t=2}^{\tau} \frac{P_{X_{t-1}X_t}^{(1)}}{P_{X_{t-1}X_t}^{(0)}}$$

where $\mu_{P^{(i)}}$ represents the initial operational state distribution under hypothesis H_i .

- **Recovering Additivity via the Logarithm:** By shifting to log-space, the multiplicative process transforms into a beautifully additive sequence of sequential step transition actions:

$$\log L_\tau = \log \frac{\mu_{P^{(1)}}(X_1)}{\mu_{P^{(0)}}(X_1)} + \sum_{t=2}^{\tau} \log \frac{P_{X_{t-1}X_t}^{(1)}}{P_{X_{t-1}X_t}^{(0)}}$$

Extending the Lower Bound: The Poisson Equation

- **The Factorization Identity Outcome:** Summing this system up to the stopping time τ leaves behind a deterministic scale and a zero-mean martingale difference sequence:

$$\mathbb{E}_{H_1}[\log L_\tau] = \underbrace{\mathbb{E}_{H_1}[\tau] \cdot \sum_i \sum_j \rho_i^{(1)} P_{ij}^{(1)}}_{\text{deterministic scale}} \log \left(\frac{P_{ij}^{(1)}}{P_{ij}^{(0)}} \right)$$

The conditional KL divergence is weighted over the unique stationary distribution $\rho^{(1)}$ of the chain under H_1 .

Moustakides Martingale

Moustakides G. V. (1999). "Extension of Wald's first lemma to Markov processes." *Journal of Applied Probability*, 36(1), 48–59.

The Markov SPRT

Algorithm 2 Markov SPRT

- 1: **Input:** Target error thresholds δ
- 2: **Initialize boundaries:** $A = \log \left(\frac{1}{\delta} \right)$
- 3: Observe starting state X_1 , set count $n \leftarrow 1$, and baseline walk $L_1 \leftarrow 0$
- 4: **while** true **do**
- 5: $n \leftarrow n + 1$, Observe next sequential state X_n
- 6: Update the trajectory log-likelihood step ratio:

$$L_n = L_{n-1} + \log \left(\frac{P^{(1)} X_{n-1} X_n}{P^{(0)} X_{n-1} X_n} \right)$$

- 7: **if** $L_n \geq A$ **then**
- 8: **Stop** and declare H_1
- 9: **end if**
- 10: **if** $L_n \leq -A$ **then**
- 11: **Stop** and declare H_0
- 12: **end if**
- 13: **end while**

Markov SPRT: δ -PAC Guarantee and the Ergodic Theorem

Performance of the Markov SPRT

Assuming H_1 is true, the expected sample path satisfies:

$$\mathbb{E}_{H_1} [\tau_{\text{Markov-SPRT}}] \leq \frac{\log(1/\delta)}{D_{\text{KL}}(P^{(1)} \parallel P^{(0)})} (1 + o(1)),$$

where

$$D_{\text{KL}}(P^{(1)} \parallel P^{(0)}) = \sum_i \sum_j \rho_i^{(1)} P_{ij}^{(1)} \log \left(\frac{P_{ij}^{(1)}}{P_{ij}^{(0)}} \right)$$

is the **stationary-weighted KL divergence rate** between the two transition laws.

Theoretical Properties:

1. **δ -PAC Classification:** Setting the threshold $A = \log(1/\delta)$ controls Type-I/II error even though the data is Markov.
2. **Where the Denominator Comes From:** From the **ergodic theorem for Markov chains**: for an ergodic chain with stationary distribution $\rho^{(1)}$

Asymptotic Optimality in the Markov Case

- **Step 1: The Asymptotic Lower Bound (for any $\pi \in \Pi_\delta$)**

$$\mathbb{E}_{H_1}[\tau] \geq \frac{d(\delta, 1 - \delta)}{D_{\text{KL}}(P^{(1)} \parallel P^{(0)})} \xrightarrow{\delta \rightarrow 0} \frac{\log(1/\delta)}{D_{\text{KL}}(P^{(1)} \parallel P^{(0)})} (1 - o(1))$$

- **Step 2: The Algorithmic Upper Bound (Markov SPRT Performance)**

$$\mathbb{E}_{H_1}[\tau_{\text{Markov-SPRT}}] \leq \frac{\log(1/\delta)}{D_{\text{KL}}(P^{(1)} \parallel P^{(0)})} (1 + o(1))$$

- **Step 3: The Asymptotic Convergence as $\delta \rightarrow 0$**

$$\lim_{\delta \rightarrow 0} \frac{\mathbb{E}_{H_1}[\tau_{\text{Markov-SPRT}}]}{\log(1/\delta)} = \frac{1}{D_{\text{KL}}(P^{(1)} \parallel P^{(0)})}$$

Foundational Bound: Sethi A., Sagar K. S., Agrawal S., Basu D., and Karthik P. N. (2026). "Asymptotically Optimal Sequential Testing with Markovian Data." *ICML 2026 Main Conference on Hypothesis Testing*, Seoul, South Korea. (Thm. 4.1, specialized to singleton null/alternative.)

Hidden Markov Models: Setup

An **HMM** is a bivariate stochastic process $\{(X_t, Y_t)\}_{t \geq 1}$ with two layers:

Hidden Layer

$\{X_t\}$ is a Markov chain on a finite state space $\mathcal{S} = \{1, \dots, d\}$ with transition probability matrix (TPM):

$$\mathbf{P} = (p_{ij}), \quad p_{ij} = P(X_{t+1} = j \mid X_t = i)$$

Observation Layer

Each observation Y_t is generated from the current hidden state X_t via an **emission function** f :

$$Y_t = f(X_t, \varepsilon_t)$$

where ε_t captures additional randomness. The emission need not be one-to-one

Detection of Stealthy Jamming using HMM

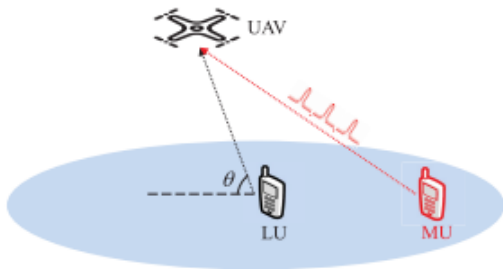


Fig. 1. The considered UAV-assisted wireless communication system under RSPJ.

What is X_m ? (The Hidden State)

- $X_m \in \{0, 1\}$ represents the jammer's behavior.
- 0 = Silent , 1 = Jammer is Active.

What is Y_m ? (The Observation)

- $Y_m = 10 \log_{10} \left(\sum |r(n)|^2 \right)$.
- The logarithmic, sliding-window RF energy captured by the UAV antenna.

The Hypothesis Family Group

- $\mathcal{H}_0$ (Normal): No attacker.
- $\mathcal{H}_1$ (Attack): A stealthy jammer actively transitions dynamically between $0 \leftrightarrow 1$.

Dynamics, Emissions & System Knowledge

1. Transition Dynamics (How the State Shifts)

- The temporal habits of the hidden attacker are driven by matrix G :

$$G = \begin{bmatrix} g_{00} & g_{01} \\ g_{10} & g_{11} \end{bmatrix} \quad \text{where } g_{ik} = \Pr(X_{m+1} = k \mid X_m = i)$$

2. Emission Mechanics (How the Math Manifests)

- Conditioned on $X_m = i$, the decibel energy follows a Gaussian distribution as we choose the window size to be greater than 30 and CLT kicks in:

$$f(Y_m = o_m \mid X_m = i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(o_m - \mu_i)^2}{2\sigma_i^2}\right)$$

- Under attack ($X_m = 1$), power shifts higher ($\mu_1 > \mu_0$) and becomes structurally chaotic and erratic ($\sigma_1^2 > \sigma_0^2$).

Reference Paper

C. Zhang, L. Zhang, T. Mao, Z. Xiao, Z. Han and X. -G. Xia, "Detection of Stealthy Jamming for UAV-Assisted Wireless Communications: An HMM-Based Method," *IEEE Transactions on Cognitive Communications and Networking*, vol. 9, no. 3, pp. 779-792, June 2023.

Application: Change-Point Detection in State Space Models

State-Space Model

$$X_n = F_\theta(X_{n-1}, \eta_n), \quad Y_n = G_\theta(X_n, \varepsilon_n),$$

where

- X_n : hidden system state,
- Y_n : noisy observation,
- θ : system parameter,
- $F_\theta(\cdot)$ and $G_\theta(\cdot)$ are known.

Change-Point Model

$$\theta = \begin{cases} \theta_0, & n < \omega, \\ \theta_1, & n \geq \omega, \end{cases}$$

where ω is an unknown change time.

Observations follow

$$Y_1, \dots, Y_{\omega-1} \sim P_{\theta_0}, \quad Y_{\omega}, Y_{\omega+1}, \dots \sim P_{\theta_1}.$$

Objective

Using only the noisy observations $\{Y_n\}$,

detect ω

as quickly as possible after the parameter changes, while controlling the false-alarm rate.

Reference: C.-D. Fuh, Asymptotically Optimal Change Point Detection for Composite Hypothesis in State Space Models, IEEE Transactions on Information Theory, 2021.

Why HMMs Are Hard: The Likelihood

Setup: We observe $Y_1, \dots, Y_n$ only. Hidden states $X_1, \dots, X_n$ are never seen.

To write the likelihood, we must **sum over all possible hidden paths**:

$$P(Y_1 = y_1, \dots, Y_n = y_n) = \sum_{x_1} \cdots \sum_{x_n} \mu_{x_1} \cdot b_{x_1}(y_1) \cdot p_{x_1 x_2} \cdot b_{x_2}(y_2) \cdots p_{x_{n-1} x_n} \cdot b_{x_n}(y_n)$$

where $b_i(y) = P(Y_t = y \mid X_t = i)$ is the emission probability.

This is a sum over $|\mathcal{S}|^n$ terms — exponential in n .

Now take the log:

$$\log P(y_1, \dots, y_n) = \log \sum_{\text{all paths}} [\text{product along path}]$$

The core obstacle

Log of a sum does not simplify. Unlike the i.i.d. and Markov cases, there are no additive increments. ℓ_n is not a random walk. Wald's theory does not apply directly.

The Original Intuition

Observation: $\{Y_t\}$ is driven by a Markov chain $\{X_t\}$.

Intuition: Maybe $\{Y_t\}$ is *also* Markov — just with a different TPM $\mathbf{Q}$?

If so, the plan was clean:



This would reduce the HMM testing problem to a **standard Markov testing problem** on the observed process — which is already solved.

Key Question

Is a function of a Markov chain itself Markov always?

The Plan Fails: $\{Y_t\}$ Need Not Be Markov

Lumpability theory tells us that $\{Y_t\}$ inherits the Markov property from $\{X_t\}$ only under **special conditions** and not in general.

Counterexample : $\mathcal{S} = \{1, 2, 3\}$, $\mathcal{Y} = \{1, 2\}$, emission map

$$f(1) = 1, \quad f(2) = f(3) = 2$$

$$\mathbf{P} = \begin{pmatrix} 1-p & p & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad 0 < p \leq 1$$

Check the Markov property for $\{Y_t\}$: We need

$$\mathbb{P}(Y_{n+1} = m \mid Y_n) = \mathbb{P}(Y_{n+1} = m \mid Y_n, Y_{n-1}, \dots)$$

Lumping Need Not Preserve the Markov Property

Kemeny J. G. and Snell J. L. (1976). *Finite Markov Chains*. Springer-Verlag, New York. (Ch. 6, on lumpability.)

Is $\{Y_t\}$ Markov? — Sequence 1

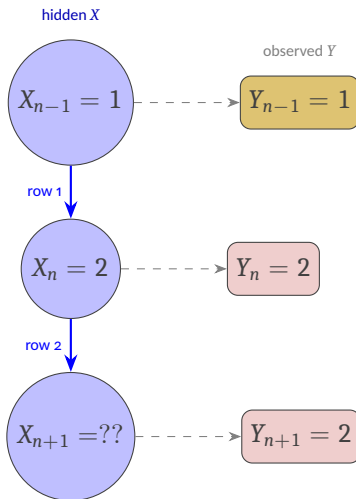
$$\mathbf{P} = \begin{pmatrix} 1-p & p & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$f: 1 \mapsto 1, \quad 2, 3 \mapsto 2$$

History: $Y_{n-1} = 1, Y_n = 2$

$$\mathbb{P}(Y_{n+1} = 1 \mid Y_n = 2, Y_{n-1} = 1) = 0$$

X never returns to 1 next step.



Is $\{Y_t\}$ Markov? — Sequence 2

$$\mathbf{P} = \begin{pmatrix} 1-p & p & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

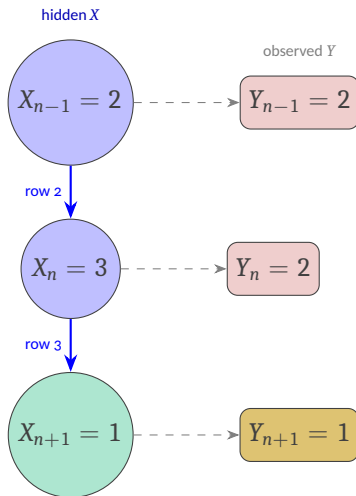
$$f: 1 \mapsto B(1), \quad 2, 3 \mapsto B(2)$$

History: $Y_{n-1} = 2, Y_n = 2$

$$\mathbb{P}(Y_{n+1} = 1 \mid Y_n = 2, Y_{n-1} = 2) = 1$$

Same Y_n , opposite outcome!

$\{Y_t\}$ is **not** Markov.



Composite vs. Composite In Markov

We now move from testing two *known* matrices to testing two *sets* of matrices:

$$H_0 : P \in \mathcal{P} \quad \text{vs.} \quad H_1 : P \in \mathcal{Q}, \quad \mathcal{P} \cap \mathcal{Q} = \emptyset.$$

Why This Is Hard

- No single reference matrix to compare against — the null itself must be *searched over*.
- The test must rule out *every* $P \in \mathcal{P}$ simultaneously, while controlling error uniformly over $\mathcal{Q}$.
- The sequential test is driven by the empirical transition matrix

$$\hat{Q}_t(x, y) = \frac{N_{x,y}}{N_x},$$

Asymptotically Optimal Sequential Testing with Markovian Data

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Abstract

We study one-sided and α -correct sequential hypothesis testing for data generated by an ergodic, finite-state Markov chain. The *null* hypothesis is that the unknown transition matrix belongs to a prescribed set $\mathcal{P}$ of stochastic matrices, and the *alternative* corresponds to a disjoint set $\mathcal{Q}$. We establish a *non-asymptotic instance-dependent lower bound* on the expected stopping time of any valid sequential test under the alternative, which is asymptotically tight. Our novel analysis improves the existing lower bounds, which are either asymptotic or provably sub-optimal in this setting. Our lower bound incorporates both the stationary distribution and the transition structure induced by the unknown Markov chain. We further propose an optimal test whose expected stopping time matches this lower bound asymptotically as $\alpha \rightarrow 0$. We illustrate the usefulness of our framework through applications to sequential detection of model misspecification in Markov Chain Monte Carlo and to testing structural properties, such as the linearity of transition dynamics, in Markov decision processes. Our findings yield a sharp and general characterization of optimal

theoretically subtle (Phatarfod, 1965; Györfi & Paulin, 2015; Fausß et al., 2020).

A widely used model of dependence is *Markovianity*, where the future is conditionally independent of the past given the present state (Bengio et al., 1999; Nagaraj et al., 2020). Markov dynamics arise in Markov Chain Monte Carlo (MCMC) (Roy, 2020), reinforcement learning and control (Sutton et al., 1998; Bertsekas, 2017; Beutler & Ross, 1985), and hidden Markov models (Rabiner & Juang, 2003). In these settings, the transition mechanism is typically unknown; theory often proceeds by imposing structural assumptions (e.g., Gaussianity or bilinearity) (Jin et al., 2020; Ouhamma et al., 2023), whose validity may be unclear. This motivates *testing* model classes under Markovian data (Natarajan, 2003; Fausß et al., 2020).

We study *sequential* hypothesis testing for finite-state Markov chains. Let $[m] := \{1, \dots, m\}$, Δ_m be the simplex in $\mathbb{R}^m$, and $\mathcal{M}$ the set of $m \times m$ row-stochastic matrices. A time-homogeneous Markov chain is specified by (P, μ) with $P \in \mathcal{M}$ and $\mu \in \Delta_m$, generating $X_1, X_2, \dots$ via

$$\mathbb{P}_{(P, \mu)}[X_1 = x_1, \dots, X_n = x_n] = \mu(x_1) \prod_{t=2}^n P(x_t | x_{t-1})$$

SPRT in HMM: Simple vs Simple

Hidden state — finite state space:

$$X_n \in \mathcal{X} = \{1, \dots, d\}, \quad n = 0, 1, 2, \dots$$

Observation — continuous, real-valued:

$$Y_n \in \mathbb{R}, \quad n = 0, 1, 2, \dots$$

X_n is never seen; only Y_n is.

The Parameter θ Drives Both Laws

θ is a single parameter (vector) that simultaneously controls:

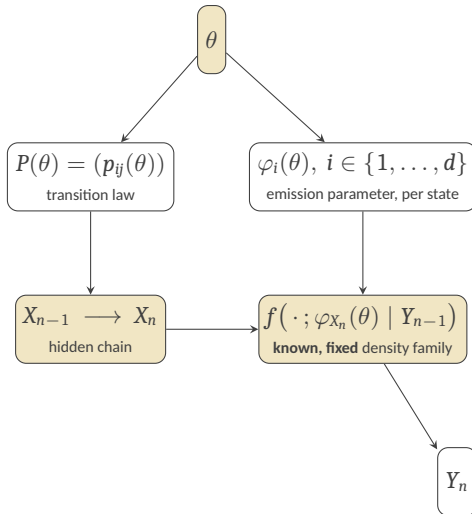
- the **transition law** of the hidden chain:

$$P(\theta) = (p_{ij}(\theta))_{i,j \in \mathcal{X}}, \quad p_{ij}(\theta) = \Pr(X_{n+1} = j \mid X_n = i)$$

- the **emission parameter** for each hidden state:

$$\varphi_i(\theta), \quad i \in \{1, \dots, d\}$$

From θ to Y_n : the Chain of Dependence



Testing θ_0 vs. θ_1 , and the Two Optimality Results

Hypotheses — two fully specified, known competitors:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta = \theta_1$$

Since $P(\theta)$ and f are known functions, plugging in θ_0 or θ_1 gives two fully pinned-down models, so the log-likelihood ratio $\log S_n$ can be computed directly from the data.

- A **lower bound** for the testing problem is established
- The **SPRT** is applied to the hidden Markov model and shown to **asymptotically attain** this lower bound

C.-D. Fuh, *SPRT and CUSUM in Hidden Markov Models*, Annals of Statistics, 2003.

Future Work

1. **Complete understanding of Fuh (2003).** Work through the full proofs in the Appendix — products of Markov random matrices, the spectral/perturbation machinery (Nagaev, Bougerol), and the Markov renewal arguments underlying the corrected Brownian approximations.
2. **Composite hypotheses via best-arm identification.** Extend from simple $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$ to θ ranging over *families* (e.g. $\theta \leq \theta_0$ vs. $\theta > \theta_0$ as a genuine composite problem), and reformulate this as a **best-arm identification** problem with HMM-generated rewards
3. **Simulation study.** Implement the SPRT synthetic HMMs (e.g. Examples 1–3 of Section 6 in FUH 2003) .
4. **Unknown emission function.** Relax the assumption that f is known.

Collaborators



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Thank You!



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The Original Intuition

Observation: $\{Y_t\}$ is driven by a Markov chain $\{X_t\}$.

Intuition: Maybe $\{Y_t\}$ is *also* Markov — just with a different TPM $\mathbf{Q}$?

If so, the plan was clean:



This would reduce the HMM testing problem to a **standard Markov testing problem** on the observed process — which is already solved.

Key Question

Is a function of a Markov chain itself Markov always?

The Plan Fails: $\{Y_t\}$ Need Not Be Markov

Lumpability theory tells us that $\{Y_t\}$ inherits the Markov property from $\{X_t\}$ only under **special conditions** and not in general.

Counterexample : $\mathcal{S} = \{1, 2, 3\}$, $\mathcal{Y} = \{1, 2\}$, emission map

$$f(1) = 1, \quad f(2) = f(3) = 2$$

$$\mathbf{P} = \begin{pmatrix} 1-p & p & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad 0 < p \leq 1$$

Check the Markov property for $\{Y_t\}$: We need

$$\mathbb{P}(Y_{n+1} = m \mid Y_n) = \mathbb{P}(Y_{n+1} = m \mid Y_n, Y_{n-1}, \dots)$$

Lumping Need Not Preserve the Markov Property

Is $\{Y_t\}$ Markov? — Sequence 1

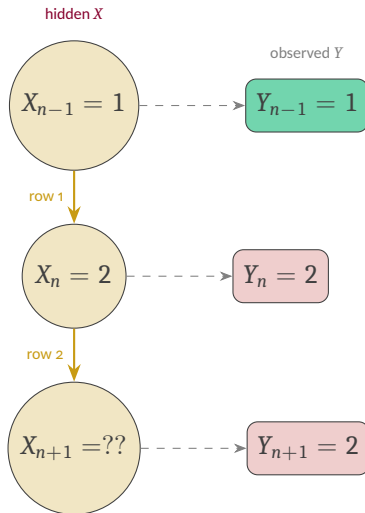
$$\mathbf{P} = \begin{pmatrix} 1-p & p & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$f: 1 \mapsto 1, \quad 2, 3 \mapsto 2$$

History: $Y_{n-1} = 1, Y_n = 2$

$$\mathbb{P}(Y_{n+1} = 1 \mid Y_n = 2, Y_{n-1} = 1) = 0$$

X never returns to 1 next step.



Is $\{Y_t\}$ Markov? — Sequence 2

$$\mathbf{P} = \begin{pmatrix} 1-p & p & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$f: 1 \mapsto B(1), \quad 2, 3 \mapsto B(2)$$

History: $Y_{n-1} = 2, Y_n = 2$

$$\mathbb{P}(Y_{n+1} = 1 \mid Y_n = 2, Y_{n-1} = 2) = \mathbf{1}$$

Same Y_n , opposite outcome!

$\{Y_t\}$ is not Markov.

